Date

Lab #05

Name : ZAKIA bibi

Class : BSCS

Section : E

lab Description

A for loop is a repetition control structure that allows us to execute a statement or a group of statements a specific number of times.

Syntax of for loop:

```
for (initialization; Condition; increment)
{
    statement;
}
```



Example

```
#include <iostream std>
using namespace std;
int main()
 {
   for (int count = 0; count < 5; count++)
   {
     Cout << "Hello" << endl;
   }
    return 0;
 }
```

Output

```
Hello
Hello
Hello
Hello
Hello
```

Switch Case:

The switch... case statement is a branching statement used to perform action based on different choices. it is an alternative to using

Date

multiple if else statement in suitable.

Syntax:

```
Switch (expression)
{
  case1:
      Statement;
      break;
  Case 2:
      Statement;
      break;
  default:
      Statement;
}
```

Task 1:

```
# include < iostream >
using namespace std;
int main ()
{int num;
 Cout<< "Enter a number:";
 Cin>> num;
 for (int i = 1;  i <= 10;  i++)
    {
     cout << num << "*" << i
      << "=" <<num *i << endl;
```

Date

```
    }
        return 0;
}
```

out put

5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Task 2:

```
#include<iostream>
using namespace std;
int main()
{
    int base, exponent;
    int power = 1; => cout << "Enter the base:";
                      cin >> base;
    cout << "Enter the exponent:";
    cin >> exponent;
    for (int i=1; i<= exponent; i++)
```

Date

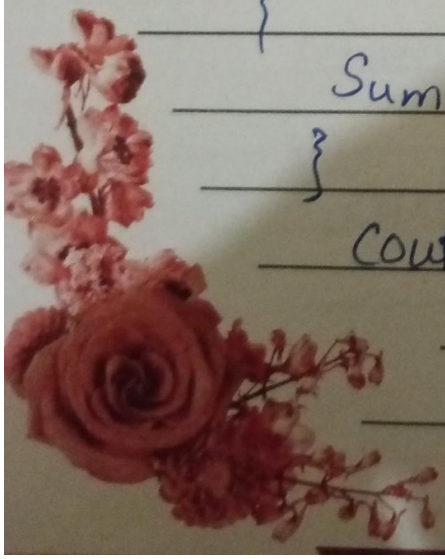
```
{
    Power = power * base;
}
Cout << "power = " << power << endl;
return 0;
}
```

Out put

Enter the base: 2
Enter the exponent. 3
Power = 8

Task 3:

```
#include<iostrem>
using namespace std;
int main () {
    int n, Sum = 0;
    Cout << "Enter a number.";
    Cin >> n;
    for (int i = 1; i <= n; i++)
    {
        Sum = Sum + i;
    }
    Cout << "Sum = " << Sum << endl;
    return 0;
    }
```



Date

Out put:

Enter a number. 5

Sum = 15

Task 4:

```
#include<iostream>
using namespace std;
int main ()
{
  int num, digit, Sum = 0;
  Cout << "Enter a number:";
  Cin >> num;
  While (num >0)
  {
    digit = num % 10;
    Sum = Sum + digit;
    num = num / 10;
  }
  Cout << "Sum of digits = " << Sum << endl;
  return 0;
  }
```

Out put

Enter a number. 1234

Sum of digits = 10

Task 1: Multiplication table

```cpp
#include <iostream>
using namespace std;
int main()
{
int num;
cout<<"Enter a number:";
cin>>num;
for(int i=1;i<10;i++)
{
    cout<<num<<"*"<<i<<"="<<num*i
<<endl;
    }
    return 0;
}
```

Task 2: Power of a number

```cpp
#include<iostream>
using namespace std;
int main()
{
int base, exponent;
int power=1;
cout<<"Enter the base:";
cin>>base;
cout<<"Enter the exponent:";
cin>>exponent;
for(int i=1; i<=exponent;i++)
{
    power=power*base;
    }
    cout<<"power="<<power<<endl;
    return 0;
}
```

Task 3: Sum of 1 to n

```
#include<iostream>
using namespace std;
int main()
{
    int n, sum=0;
    cout<<"Enter a number:";
    cin>>n;
    for(int i=1; i<=n;i++)
    {
        sum=sum+i;
    }
    cout<<"sum="<<sum<<endl;
    return 0;
}
```

Task 4: Sum of digits

```
#include<iostream>
using namespace std;
int main()
{
int num,digit,sum=0;
cout<<"Enter a number:";
cin>>num;
while(num>0)
{
    digit=num%10;
    sum=sum+digit;
    num=num/10;
    }
    cout<<"sum of
digits="<<sum<<endl;
    return 0;
}
```